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Societal Impact. Mechanistic interpretability is focused on understanding the key mechanisms at play in deep learning systems by tracing them down to the weights. It is often associated with AI safety, hoping that a deeper understanding of these systems can help us steer them to be more "aligned" with "human values", and prevent AI dystopia scenarios. On the other hand, it could also prove useful in training more powerful models, which is associated with significant societal issues linked to the rise of advanced AI systems. These issues are too broad to be discussed in this paragraph.

A Additional Figures & Findings

Figure 11 studies the probability of finding the iteration head when learning with the parity data. As mentioned in the main text, it showcases the bigger probability of learning other circuits when learning the parity problem only.

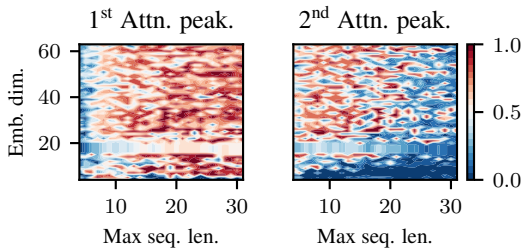

Figure 11: Same figure as Figure 8 yet when learning with the parity dataset. The whitening of the row around $d = 20$ is due to GPU failure, and should not be considered when parsing this figure.

Figure 12 showcases the usefulness of large learning rates and small batch size when using SGD, and the correction brought by Adam.

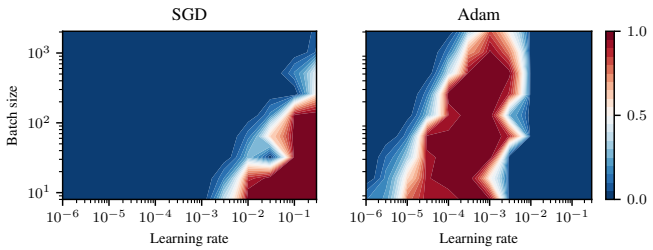

Figure 12: Test accuracy for SGD and Adam after 100 epochs.

Finally, Figure 13 plots some attention maps when learning with a three layers transformer with two attention heads per layer. Knowing the iteration head circuit, we are able to observe a similar circuit, yet with work shared across heads and layers.

Position embeddings. When we started this project, we were expecting to find some grokking structure emerged from the need to perform position subtraction. In particular, as mentionned in the main text, we were expecting this mechanism to appear as the position embedding dimension was small. When learning on the parity problem, we found that the network was not implementing the iteration head when the position embedding dimension was really small. This can notably be seen on Figures 15 and 14. We notably observed that freezing the position embedding does not change much the picture, which can be seen as a result of overparameterization. Similar type of observation were observed when learning with the polynomial iteration problem, as reported on Figure 16.

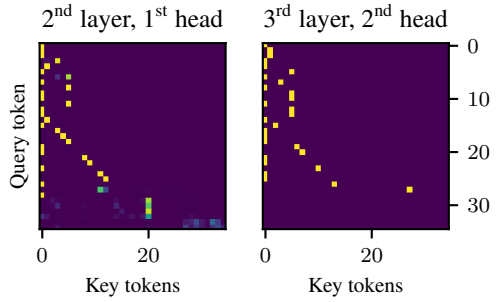

Figure 13: Recovering the "who is p_t?" key-query association, yet shared across layers and heads when training a three layers transformer with two attention heads per layer.

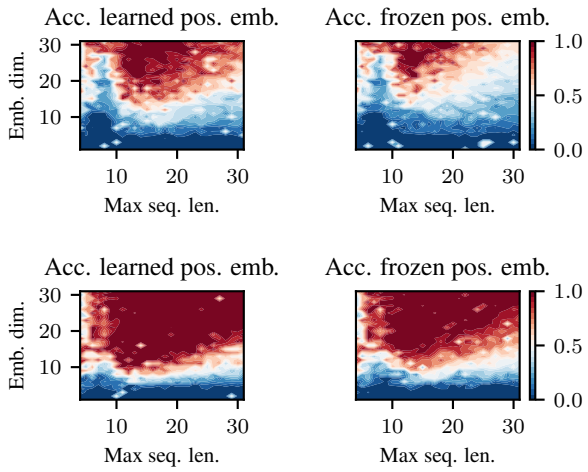

Figure 14: The effect of small embeddings when learning the parity problem. The top row corresponds to what has been learned after 1000 epochs. The bottom one corresponds to 5000 epochs.

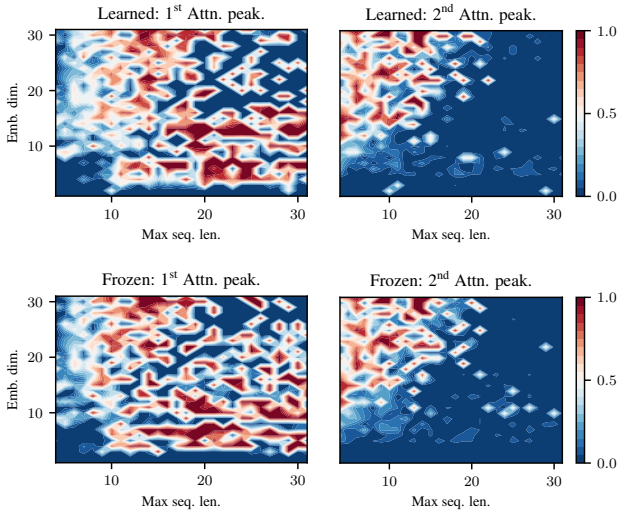

Figure 15: Attention learned when studying frozen vs learned positional embedding.